The Living Economy

A complete design for the economy of Rise of Mankind: land, people, food, goods, living standards, industry, markets, trade, the state and war, as one system

This document describes the economy as it should be, whole, rather than a sequence of patches. It replaces the map's resource tokens, the on/off resources, the leasing of resources for gold, and the fixed yields of specialists with one ledger of goods counted in whole tons. That ledger is made by people working land and workshops, eaten and used by the population, transformed by industry, priced by markets, carried by trade and taxed by the state. Food works as the engine's food works today: a city eats first and grows from what is left; what is new is that the food it keeps can rot, and how much rots depends on the technologies it lacks. The game's three yields and four commerce outputs remain what the player sees, because the executable's screens show exactly those; underneath, they become views of the ledger.

What it rests on. Every fact about today's game is from the mod's XML and DLL at GitLab `main` `3bec049` (the XML is unchanged at `ae60523`). Every measured number is from *Land, not tokens*, *Settlement Economy* and the growth runs of this document (research/economy): 30 real harness games, rebuilt city by city, and deterministic models run on them. Every parameter is an anchor taken from the engine, a quantity solved from those measurements, or a design value stated as such, with the measured consequence of changing it (section 18).

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1. Playing it

4000 BC. Your first village of three eats 900 t of grain and fish a turn and harvests 1,300 t. Before Pottery, the surplus sits in baskets and pits, and a share of it rots every turn: the ledger shows the food box filling, then slowing, then barely moving. Across the 30 measured games, cities that never learnt Pottery would have lost 17 % of their harvest to spoilage. Pottery is one technology away. You research it, the box fills, and the village grows.

2000 BC. Your capital sits where a river crosses warm plains. The overlay shows why the site was good: wheat suits the flood plain at 0.8, the hills to the north hold copper of grade 0.7, the coast is shelf water. The town of size 4 has two farmers, one herder and one fisher; it eats 1,200 t a turn and grows from what is left, as cities always did. Its people also want meat or fruit, wool or cloth, and firewood. The ledger reads: *Clothing: 30 t of 60 t; 2 of 4 citizens clothed*. A neighbour offers wool for your copper. You accept a contract: 30 t of wool a turn for 20 t of copper a turn, for 20 turns. Fermentation came last century: the fish is salted in jars and the fruit pressed into wine, so very little of either rots any more.

400 AD. Eleven cities, one network of roads and a river. The market shows grain cheap and wine dear; your coastal cities are fed by the inland farms, which is why the iron town in the hills could grow to 9 on rock. Three of your farm towns are as large as their health and contentment allow; their surplus, 2,988 t a turn between them, goes down the river to the capital and out to sea under a grain contract. A merchant from the south sells you rice seed for 300 gold, and the river delta that was marsh becomes paddy over the next ten turns.

1850. Coal and iron meet in the north; four mills turn them into steel, and the steel into rails and rifles. A third of your people now work in workshops and factories rather than fields, fed by grain bought abroad under a 50-turn contract. Refrigeration is still a generation away, so the beef from the pampas comes salted and the cities' milk comes from their own hinterland. Your rival blockades the strait; the price of cotton doubles in a turn, two mills stand idle, and contentment in the textile towns falls. You can pay, fight, or plant cotton in the south, which the climate overlay says will grow at 0.4.

Every one of these situations follows from the rules below. None needs a scripted event.

2. Pillars

Pillar	Meaning	Consequence for the design
Material truth	Things are made of things. Nothing appears from nowhere or is used without being made	One conserved ledger in whole tons; recipes consume inputs; armies cost goods
The land matters	Where you are decides what you can do easily, and what you must trade for	No tokens; climate, soil, ore and sea as qualities of every plot
People are the economy	Output is work; what people need decides what they do and how content they are	Occupations; living standards counted in whole citizens; needs grow with size
Feeding comes first	A city eats before anything else; hunger is a crisis, not a background state	The engine's growth rule kept; spoilage only where technology is missing
Legibility	Every number can be traced to its cause in two clicks, and reads as a quantity	Tons and citizens on every screen; never a fraction or a ratio
Consequence without micromanagement	Choices are about direction, not clicks	Automatic distribution within networks; governors that know needs; contracts that run themselves
AI parity	The AI plays the same economy and understands it	One valuation engine for every AI decision (section 16)
Determinism	Every computer reaches the same result, bit for bit	Whole tons in integers sized so they cannot overflow; exact rationing; the game's synchronised random numbers

3. The whole system

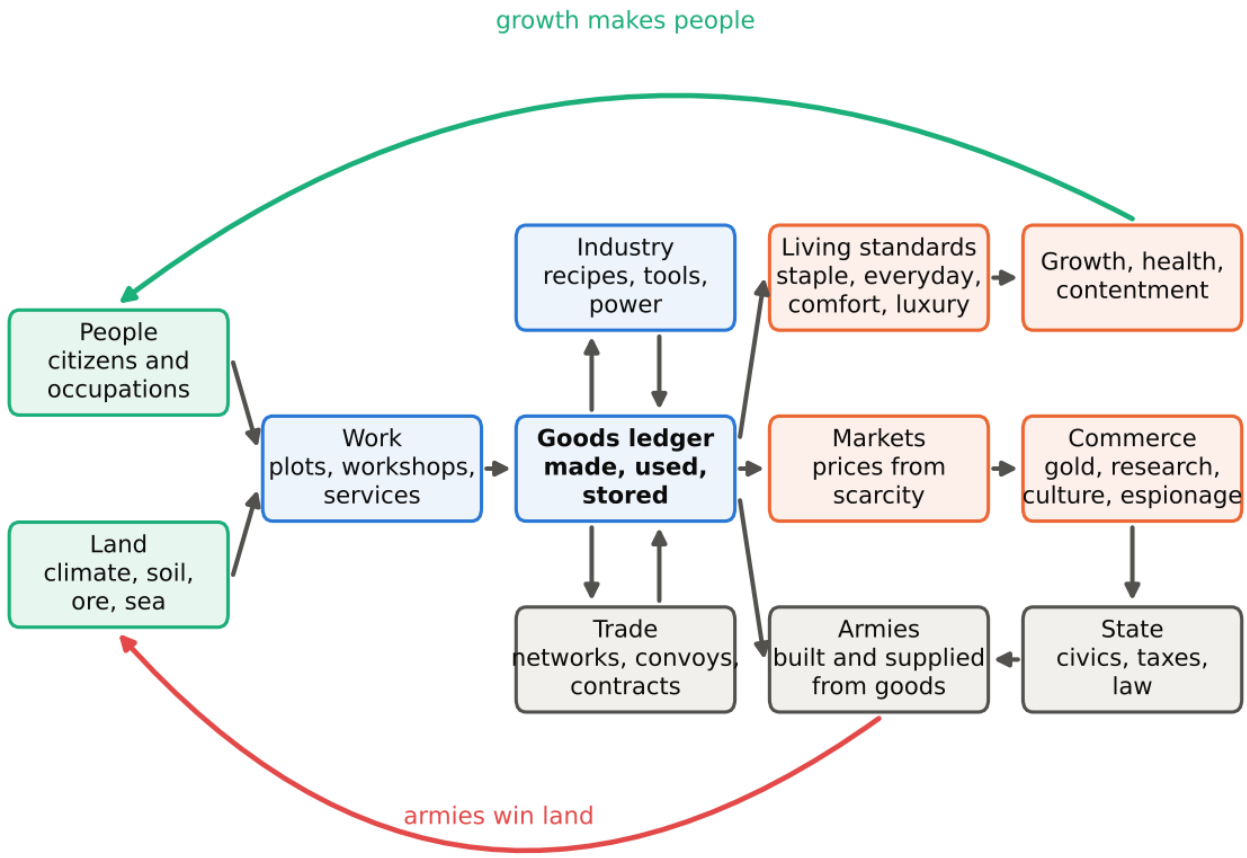


Figure 1. The economy as one system. Land and people produce work; work fills the goods ledger; goods feed living standards, industry, markets, trade and armies; markets produce commerce for the state; growth makes people; armies win land; policy shapes work.

There are three loops, and every mechanic belongs to at least one. The **loop of life**: land and work make food and everyday goods; citizens eat, and the surplus makes the city grow, which makes more work. The **loop of wealth**: surplus goods are sold in markets, commerce is taxed into gold, research and culture, which buy buildings and technology that make work more productive. The **loop of power**: goods make and supply armies, armies take or protect land and trade, and land and trade feed the other two loops.

The design's central measured fact is in figure 2: in the 30 games, the worked plots account for nearly all production at first and for about a quarter of it by turn 150 (production 4.08 times what the plots give, middle half 3.1 to 5.1). A game of Rise of Mankind is the passage from an economy of land to an economy of capital. The Living Economy makes that passage visible and material: the same share that is a modifier today becomes workshops, tools, power and the goods they consume.

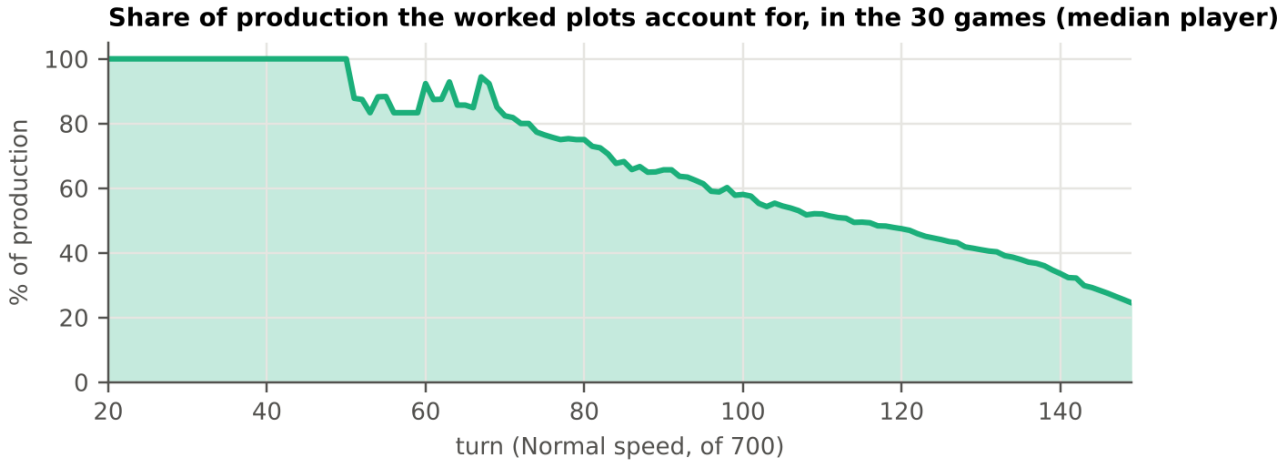


Figure 2. The share of production that the worked plots account for, as the median player's plots' production divided by its observed production (CvCity::getYieldRate), in the 30 games. The rest comes from buildings, civics and specialists.

4. What is cut, replaced and kept

Today	In the Living Economy	Why
Resource tokens on plots; improvements that unlock them	Cut. Land qualities: crop suitability, ore grade and depth, fishing grounds	Places get character; resources follow geography (Land, not tokens)
A resource is on or off for a network	Cut. Goods are quantities in whole tons, made and used each turn	Scarcity, specialisation and trade become possible
+1 happiness or health per resource, whatever the city's size	Replaced. Living standards in tons per citizen, counted in whole citizens supplied	Large cities need more; luxury becomes a matter of supply
Leasing a resource copy for gold per turn	Replaced. Shipments and contracts in tons, seeds and livestock, packages	Between AIs, 568 of 580 resource deals were swaps of copies and 1 paid gold
Buildings that need resources present and give a copy	Replaced. Recipes that turn inputs into outputs, run by workers	Factories should use what they need
Specialists with fixed yields	Replaced. Occupations that produce goods (smith, weaver, miller) or services (merchant, scholar, priest)	Work is what makes goods
Trade routes worth 1.00 commerce times modifiers	Replaced. Convoys that carry goods, earning on what they carry	On this map 2708 of 3025 routes paid exactly the floor
Food box, growth threshold, starvation when the box is empty	Kept, in tons: 300 t per citizen, threshold (30 + 10 × size) × 100 t; the box can spoil	The pacing of 700 turns rests on them; a city eats before it stores
Food, production, commerce on the city screen	Kept, as views of the ledger	The executable's screens show exactly these
Gold, research, culture, espionage sliders	Kept. They split taxed commerce as today	Same
Civics, technologies, buildings	Kept and re-aimed: civics become economic levers; Pottery, Fermentation and Refrigeration stop spoilage; buildings become capital	The tree and the tech web stay; their effects become material

5. The ledger in whole tons

5.1 The unit

Every good is counted in whole tons (t). One unit of today's food yield is 100 t, so a citizen eats 300 t a turn (FOOD_CONSUMPTION_PER_POPULATION = 3) and a city of size 4 grows when its box holds (30 + 10 × 4) × 100 = 7,000 t (BASE_CITY_GROWTH_THRESHOLD = 30, CITY_GROWTH_MULTIPLIER = 10, Normal speed 100 %). There are no fractions anywhere: every rule that divides rounds down, and the remainder stays where it was, so rounding neither creates nor destroys a ton. A quantity on any screen or in any deal is a whole number of tons, and a living standard is a whole number of citizens supplied (section 10), never a ratio.

The ton is the game's unit of account, not a census. The length of a turn on Normal runs from 50 years for the first 40 turns to 3 months for the last 20 (CIV4GameSpeedInfo.xml), and the engine's population figure is 1,000 × size$^{2.8}$ (`CvCity::getRealPopulation`), so no fixed ration per turn could be a historical ration at every point of the game. 100 t per unit of food was chosen because every yield in the game today stays a whole number of tons, the smallest flows worth naming (a few tons of salt or dye) are still whole, and deals read naturally: 400 t of grain a turn for 20 turns.

5.2 The ledger

Every city keeps one ledger: for every good g, its stock X, what it made q, received m, used u and sent x this turn, in tons. Every flow in the game is a line in some ledger, so goods are conserved exactly. What is left after use loses its spoilage rate $\sigma_g(K)$, in tons per thousand, for the set K of technologies the owner knows (section 6), and is held up to the city's storage C_g:

$$X_{g,\,t+1} = \min\left(C_g,\ Y_{g,\,t} - \left\lfloor \frac{\sigma_g(K)\, Y_{g,\,t}}{1000} \right\rfloor\right), \qquad Y_{g,\,t} = X_{g,\,t} + q_{g,\,t} + m_{g,\,t} - u_{g,\,t} - x_{g,\,t} \tag{1}$$

5.3 The three yields as views

The three yields become views of the ledger, so the city screen, the growth rule, the build queue and the sliders work unchanged:

$$\text{Food} = \left\lfloor \frac{1}{100} \sum_{g \,\in\, food} (q_g + m_g) \right\rfloor \tag{2}$$

$$\text{Production} = \frac{1}{100}\, L_{work}\, \pi_0\, (1+\tau)(1+\kappa) + P_{ind} \tag{3}$$

$$\text{Commerce} = \frac{1}{100}\left(\sum_g p_g \,\text{sold}_g\, \mu + \text{services} + \text{convoys}\right) \tag{4}$$

Food is the harvest and imports of food goods in hundreds of tons, so `CvCity::foodDifference` = Food − 3 × population keeps its meaning on the city screen while the ledger holds the exact tons. Production is the work of the citizens not producing goods, times their tools τ and power κ, plus P_{ind}, what industry produces as construction capacity. Commerce is what the city's market earns from goods sold at price p with margin μ, plus services (merchants, banks) and convoys. The sliders split commerce into gold, research, culture and espionage exactly as now. The engine's percentage modifiers (`CvCity::getBaseYieldRateModifier`) stay, applied to these views, so every building and civic that changes a yield keeps working while it is re-aimed at goods.

6. Food

6.1 The rule

Food keeps the rule of `CvCity::doGrowth`, with every quantity in tons and one new term, spoilage. Each turn, in each city:

- **Harvest.** The worked plots yield food goods: grain, fish, meat and dairy, fruit. Imports arrive.
- **Eat.** Each citizen eats 300 t, taken from the harvest and the box together, the most perishable good first (fish, then meat and dairy, then fruit, then grain), so the food that keeps is the food that is kept.
- **Keep.** What is left is the food box. Each food good in it loses its spoilage rate for the technologies the owner lacks (6.2).
- **Grow.** When the box reaches the threshold, the city grows by one and keeps a share of the threshold: 20 % from Settlement Essentials (free from the ancient era), 30 % from Town Essentials (Masonry and Pottery), as today's records give.
- **Hunger.** If harvest and box together cannot feed every citizen, the box empties and the city loses one citizen, exactly as today. This is the only way a citizen goes hungry.

A city is therefore not short of food by default. It eats first; only the surplus is stored, grows the city or spoils. A city with a bad harvest draws on its box; it loses a citizen only when the box is empty.

$$\text{lost}_g = \left\lfloor \frac{\sigma_g(K)\, B_g}{1000} \right\rfloor, \qquad \sum_g B_g \geq (30 + 10\,n) \times 100 \Rightarrow n \to n+1 \tag{5}$$

6.2 Spoilage by missing technology

Spoilage depends only on what the owner does not know. The rates are tons lost per thousand held, per turn. With all three technologies, nothing rots.

Food good	No preservation	Pottery	Fermentation	Refrigeration	What stops the loss
grain	60	10	10	0	Sealed jars and granaries (Pottery)
fruit	150	100	20	0	Wine, cider, dried fruit (Fermentation)
dairy	250	250	50	0	Cheese, cured meat (Fermentation); cold store (Refrigeration)
fish	300	200	100	0	Salted and fermented fish in jars (Pottery, Fermentation); ice (Refrigeration)

The three technologies exist in RoM today, and their records place them well: **Pottery** (cost 60, ancient, 1 technology from the start for every civilization in the 30 games, which all begin with Fishing) unlocks Town Essentials, the Cottage and Tablets. **Fermentation** (cost 95, ancient, needs Agriculture, Naturopathy and Pottery) gives +1 food on farms and pastures, the Brewery, the Winery and the olive and apple farms. **Refrigeration** (cost 5,200, industrial, at least 88 technologies known) unlocks the Commercial Center (keeps 10 % of the box, +8 health) and the Cereal Mills, and +1 commerce on pastures. In the Living Economy each keeps what its record gives and adds the end of a loss: the +1 food that Fermentation gives farms today becomes the grain and fruit that no longer rot.

A box that loses a share each turn cannot grow without limit. With a steady surplus S and rate σ, it settles where the loss equals the surplus, and the city can grow only while that level is above its threshold:

$$B^{*} = \frac{S\,(1000 - \sigma)}{\sigma}, \qquad \text{growth to size } n \text{ only if } B^{*} \geq (30 + 10\,n) \times 100 \tag{6}$$

Figure 3 (left) shows one city of size 4 with 400 t of grain to spare each turn: without Pottery its box settles at 6,266 t, below the 7,000 t it needs; with Pottery it grows in 20 turns instead of 18 with no loss. A village without Pottery is therefore held small by its own surplus, not by hunger, and the larger its surplus the larger it can become.

6.3 What it does in the 30 games

The rule was run on the cities of the 30 games. Each city is founded at size 1 on its observed plot and turn (captures and razings as observed) and from then on grows only from its own food: the rebuilt plots of *Settlement Economy*, worked by the game's governor rule, with the deterministic harvest weather. Two modes bracket what else limits a city. With **food the only limit**, cities grow as fast as their land allows. With **other limits as observed**, a player's cities may not together exceed its observed population at that turn, which holds health and happiness as the games had them; a city at that limit keeps its box at the threshold, as `CvCity::doGrowth` does under avoid-growth.

Without spoilage, the first mode overshoots the games early and meets them late: 3,065 citizens against 1,153 observed at turn 50, 22,405 against 20,537 at turn 150. The rebuilt harvest leaves out the food that went into settlers and workers and the –25 % of Oral Law, which may be why it overshoots early (*open*). The second mode meets the observed population at turn 50 and falls 9.3 % short at turn 150. Spoilage is judged against the no-loss run of the same mode, so these gaps are common to every arm.

Other limits as observed (the closer bracket to the games):

Preservation known	Population at turn 50 / 100 / 150	vs no loss, turn 150	Harvest spoiled	Hunger, per 10,000 city-turns
none, all game	1,117 / 5,924 / 15,592	-16.2 %	16.9 %	0.9
Pottery, all game	1,152 / 6,209 / 18,169	-2.4 %	3.7 %	1.1
Pottery and Fermentation, all game	1,152 / 6,220 / 18,239	-2.0 %	3.0 %	1.1
each at its earliest turn	1,152 / 6,220 / 18,239	-2.0 %	3.1 %	1.1
all three (no loss, today)	1,152 / 6,244 / 18,617	+0.0 %	0.0 %	1.2

Food the only limit:

Preservation known	Population at turn 50 / 100 / 150	vs no loss, turn 150	Harvest spoiled	Hunger, per 10,000 city-turns
none, all game	2,388 / 7,906 / 16,868	-24.7 %	15.5 %	0.9
Pottery, all game	2,933 / 9,855 / 21,307	-4.9 %	3.0 %	1.8
Pottery and Fermentation, all game	2,973 / 9,973 / 21,497	-4.1 %	2.5 %	2.1
each at its earliest turn	2,920 / 9,948 / 21,489	-4.1 %	2.5 %	2.1
all three (no loss, today)	3,065 / 10,394 / 22,405	+0.0 %	0.0 %	3.3

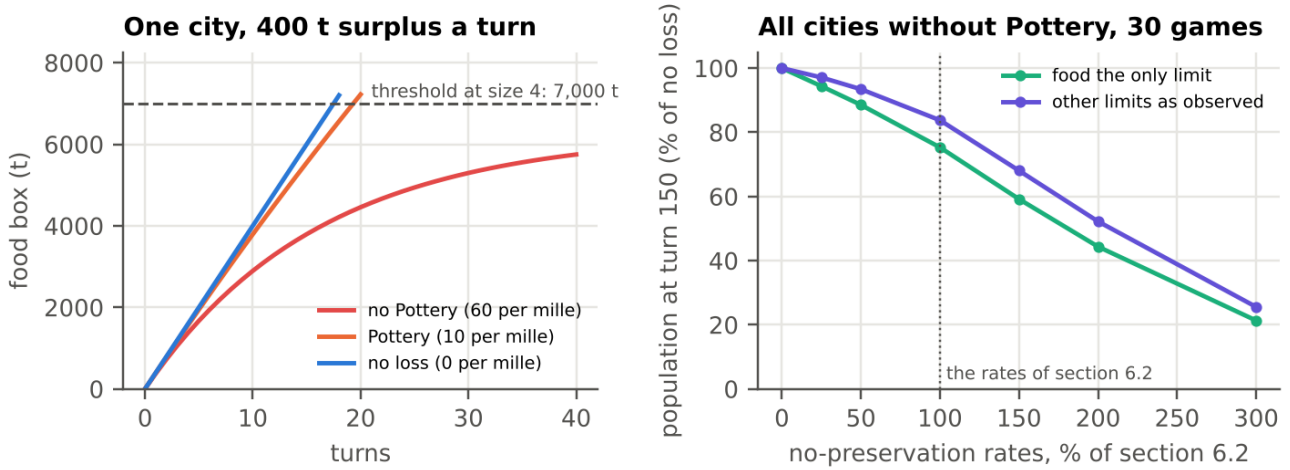


Figure 3. Left: the food box of one city of size 4 with 400 t of grain to spare each turn, under the grain rates of section 6.2; without Pottery the box settles below the threshold. Right: population at turn 150 in the 30 games with no city knowing Pottery, as the no-preservation rates are scaled from 0 to 3 times those of section 6.2, against no loss.

When the technologies arrive. Telemetry counts each player's technologies but does not name them, so the turn of Pottery is known only as a bound: the earliest turn at which a player's count had risen by the technologies it lacked. For Pottery that is turn 4 to 6 (median 6) for all 300 players; for Fermentation turn 13 to 23 (median 21). No player could have known Refrigeration by turn 150: the most advanced player of each game held 41 to 50 technologies, and Refrigeration needs at least 88. The row "each at its earliest turn" uses these bounds, so it is the most favourable case.

The food a city can sell. With other limits as observed, a city whose box reaches the threshold while its owner is at its health and happiness limit cannot grow. That happened on 44,902 city-turns, holding 996 t a time on average: 11.2 % of the whole harvest of the 30 games (each Pottery and Fermentation at its earliest turn). Today that food is lost at the threshold. In the Living Economy it is the surplus that feeds cities without farmland, fills granaries for war and sells abroad, without slowing anyone's growth.

Oral Law. Every civilization starts under Oral Law, whose record gives food –25 %. Applied to the harvest under this rule (the governor seeing it), it leaves 13,503 citizens at turn 150 instead of 18,239 (-26.0 %), with other limits as observed; hunger stays at 1.4 per 10,000 city-turns, because the governor works more food plots.

6.4 Readings

Each reading below rests on the facts named and would be contradicted by the evidence named.

- **Hunger is an event, not a state.** In every arm, a city empties its box and loses a citizen on 0.9 to 3.3 of 10,000 city-turns. The previous model counted 5.5 % of citizen-turns short of staple, because it split each player's observed population among its cities and fed each turn on its own, with no box. Contradicted if the games themselves starve far more often than the rebuilt cities: telemetry has no food per city, so this is *open*.
- **Pottery matters, early.** A civilization that never learns it ends turn 150 with 16 % fewer citizens; one that learns it and Fermentation at the earliest turns loses 2.0 % to spoilage in all. The design relies on the AI valuing Pottery by the food it saves (section 16). Contradicted if the AI, valuing it, still researches it late; its research order is not in telemetry (*open*).
- **The rates are levers, and the curve is steep.** Figure 3 (right): halving the no-preservation rates costs 7 % of turn-150 population instead of 16 %; tripling them costs 74 %. The values of section 6.2 put the cost of missing Pottery between a nuisance and a lock.
- **On a quarter of city-turns, something other than food stopped growth.** Cities held at the threshold on 44,902 of 176,331 city-turns; on the rest, food set the pace. Contradicted if the held turns came from the cap's allocation among cities rather than from real limits; the cap is per player, not per city.
- **Oral Law's food penalty cannot stay a percentage.** At –25 % of the harvest the model loses a quarter of its population, while the games reached 20,537 citizens at turn 150 starting under it. Either players leave Oral Law early or the rebuilt land is short; civic adoption is not in telemetry (*open*). Either way, Oral Law becomes a lever on what the state sees and on waste in stores (section 12.4), not on the harvest.

7. Land

Land is described in *Land, not tokens* and adopted whole: per plot, a climate (annual temperature, seasons that turn with the hemisphere, water, growing season), a soil that wears with farming and recovers, geological provinces with ore grade and three layers of reserves, and for water, a carrying capacity for fish. On the 30 games' map the reconstruction checked it against the games themselves: the rebuilt cities' worked plots feed their observed populations on 93.1 % of city-turns.

Land use (the improvement)	What it makes	Depends on
Farm (of a crop the civilization knows)	Grain or a cash crop	Crop suitability, soil, irrigation; Latifundium stages as estates
Pasture	Meat and dairy, wool, hides, horses	Grass, climate; the breed known
Orchard, vineyard, olive grove	Fruit, oil, wine	Mediterranean climate; years to bear
Plantation	Sugar, cotton, coffee, tea, tobacco, spices	Tropical climate; labour regime (section 12.4)
Mine, Shaft Mine, Modern Mine	Ores, coal, stone, salt	Province grade; depth reached; reserves left
Lumber camp, then forestry	Timber, charcoal	Forest; regrowth
Fishing fleet	Fish	Fishing ground capacity and stock
Town, then city district	Services, housing	Population; the land's commerce today

8. People and work

Every citizen has an occupation. The engine already distinguishes citizens working a plot from specialists; the Living Economy uses that distinction and fills it with meaning. RoM defines 19 specialist types today; they become the occupations below, defined in XML.

Occupation	Where	Makes
Farmer, herder, fisher, miner, forester	A worked plot	The plot's goods (section 7)
Artisan: smith, weaver, potter, miller, brewer, carpenter	A workshop slot of a building	The building's recipe, one run per artisan
Worker (from the industrial era)	A factory slot	The factory's recipe, several runs per worker with power
Merchant	A market slot	Commerce: margin on goods sold; convoy capacity
Scholar, priest, artist, official	Library, temple, theatre, court	Research, culture, contentment, administration
Labourer (no slot)	The city	Production: building and training work

The city governor assigns occupations by marginal value: each citizen goes where one more worker adds most to the city's living standard and output (section 16). In the model that measured this, a governor that knew the city's needs lowered the share of citizens short of fuel from 66.4 % to 53.4 % and of everyday food variety from 50.8 % to 43.5 % compared with one that ranked plots by yields alone.

Labour regimes are civics, and they change how work is done, not a percentage: tribal labour (everyone farms), slavery (cheap plantation and mine labour, unrest, no artisans above a size), serfdom (estates bound to land), free labour (artisans and merchants possible; wages), industrial labour (factory workers). Each sets which occupations exist, their output per worker and their effect on contentment.

9. Goods

About forty goods carry the whole game, each counted in tons. The catalogue keeps today's resource types as names wherever they fit, so that existing art, Civilopedia text and requirements can be reused.

Era of first use	Raw goods	Made goods (inputs)
Ancient	Grain (wheat, rice, maize, potato), fish, meat and dairy, fruit, wool, hides, timber, stone, clay, copper, salt	Pottery (clay, fuel), bronze (copper, tin from trade), cloth (wool), beer and wine (grain, fruit)
Classical	Iron ore, tin, marble, olives, grapes, incense, dye	Iron tools and arms (iron, charcoal), oil (olives), fine cloth (wool, dye), coin metal (silver, gold)
Medieval	Horses, silk, spices, furs	Plate armour (iron), paper and books (timber, scholars), glass (sand, fuel), furniture (timber)
Renaissance	Sugar, coffee, tea, tobacco, cotton	Gunpowder (sulphur, saltpetre), ships' stores, refined sugar, porcelain
Industrial	Coal, lead, sulphur, rubber	Steel (iron, coal), machinery (steel), rails (steel), textiles (cotton), ammunition (lead, sulphur), chemicals, tinned food
Modern	Oil, bauxite, uranium	Fuel (oil), aluminium (bauxite, power), plastics, electronics, consumer goods, fertilizer, medicine
Transhuman and future	(engineered)	Nanotubes, nanobots, personal replicators: the goods that end scarcity, as RoM's future era already names them

Rules for the catalogue. A good exists only if a choice turns on it: a different land, a different building, a different trade partner. Goods that would always travel together are one good (meat and dairy). Each era adds goods that make the previous era's goods more valuable (iron makes tools that make every farm more productive). Every made good has at least one raw good that some civilizations lack, so that trade is never optional for everyone. Every good has a spoilage rate per technology level (food in section 6.2; timber, cloth and metals 0). The catalogue size costs almost nothing to compute: 0.95 ms per turn at 300 cities and 100 goods (section 19.5).

10. Living standards

Beyond food, each citizen wants goods from three tiers, in tons per turn. A city of n citizens wanting l t each of a tier's goods and holding S t of them supplies a whole number of citizens, and the rest go without:

$$k = \min\left(n,\ \left\lfloor \frac{S}{l} \right\rfloor\right), \qquad \text{without} = n - k \tag{7}$$

The city screen says it in those words, here for a city of size 10: *Clothing: 176 t of 220 t wanted; 8 of 10 citizens clothed*. Every effect counts citizens, so it is a whole number like every other effect in the game. Tiers are nested: a citizen counts for comforts only if clothed, fed with some variety and warm.

Tier	Goods (substitutes in each line)	Per citizen supplied	Per citizen without
Food	grain · fish · meat and dairy · fruit (section 6)	Eats; the surplus grows the city	Only when the box is empty: the city loses a citizen
Everyday	meat, dairy or fruit (variety) · wool or cloth · timber, charcoal or coal	Nothing: this is the norm	Variety: 1 discontent; clothing or fuel: 1 unhealthiness, which eats 100 t of food (CvCity::foodConsumption)
Comfort	salt · wine or beer · pottery or furniture · (later) sugar, tea, coffee	1 contentment; artisans and merchants possible	Nothing lost
Luxury	spices, silk, fine cloth, dye, jewellery, porcelain	Culture and great-person points	Nothing lost: luxury is aspiration

Levels by development. A village (size 1 to 3) wants the base level per citizen; a town (4 to 8) 1.4 times it; a city (9 and more) 2.0 times. The base levels were solved, not chosen (section 18.3): they are the levels at which, in the 30 games, 8 of 10 citizens have their everyday goods and 5 of 10 their comforts. In whole tons per citizen and turn:

Need	Village (1 to 3)	Town (4 to 8)	City (9 and more)
Everyday food variety	22 t	30 t	43 t
Clothing	11 t	15 t	22 t
Fuel	6 t	8 t	12 t
Comfort	13 t	19 t	26 t

11. Production and industry

A building with a recipe has slots; each artisan in a slot runs the recipe once per turn if the inputs are there. Output follows the scarcest input (fixed proportions), and tools and power raise output per worker:

$$n_k = \min\left(\lfloor w_k\, r_k (1+\tau)(1+\kappa) \rfloor,\ \min_g \left\lfloor \frac{\alpha_{g,k} X_g}{a_{g,k}} \right\rfloor\right) \tag{8}$$

w is the number of workers, *r* runs per worker, α the building's share of a scarce input (exact proportional rationing in whole tons), *a* the tons of input per run. Tools are themselves a good that wears out: every so many worker-turns use up a ton of tools, so iron and fuel stay in demand for the whole game. Power (from the industrial era) is a network good produced by plants from fuel.

The industrial revolution is the moment workshops overtake land, which figure 2 shows happening already in the 30 games' first 150 turns. In the Living Economy it is the moment coal, iron and power let a worker run several recipes instead of one, which draws people from farms into factories and makes the empire depend on fed cities it does not feed itself. Refrigeration, at the same time, lets meat, milk and fish travel without loss, which is what lets a city outgrow its own hinterland.

12. Markets, money and the state

12.1 Markets

Each trade network has one market. After production and recipes, every surplus is offered and every unmet need bid; goods move from surplus to need inside the network automatically, in whole tons, limited by the network's transport capacity. The price of a good in the network follows its scarcity, within bounds so that no single turn can ruin a city:

$$p_g = p_g^0 \ \mathrm{clamp}\left(\left(\frac{D_g}{S_g}\right)^{1/\varepsilon_g},\ \tfrac{1}{5},\ 5\right) \tag{9}$$

Prices are shown to the player and used by the AI. Food is the largest market: the surplus held at the threshold (section 6.3, 11.2 % of the harvest in the 30 games) moves first to cities in the network that can still grow, then to stores, then abroad.

12.2 Money

Commerce is what the market earns: the margin on goods sold, plus services. The state taxes it, and the sliders split the tax into gold, research, culture and espionage as today. Coin metal (silver, gold) raises the margin a market can take; banks raise it further. Money is what buys goods across networks and from other civilizations; it is never created except by trade and taxation.

12.3 Buildings as capital

Buildings keep their roles but act on goods: the essentials buildings keep their share of the food box after growth (Settlement Essentials 20 %, Town and City Essentials 30 %, Metropolitan Essentials 40 % in today's records) and set the city's storage; a market adds merchant slots and margin; a harbour adds sea transport capacity and fishing fleets; a forge is a workshop of smiths. Of today's 389 buildings, 146 give specialist slots, which become workshop slots; 26 add trade routes, which become transport capacity.

12.4 Civics as economic levers

Civic category	Today (examples)	In the Living Economy
Administration (Oral Law, Tablets, ...)	Oral Law: food –25 %, production –25 %, commerce –10 %	How much of the ledger the state sees and taxes; waste in state stores
Economy (Hunter-Gatherer, Barter, Coinage, Planned, ...)	Trade yield modifiers; Planned: trade gives +50 % food and production	How markets clear: barter (no money, swaps only), coinage (prices), planned (the state allocates)
Employment (Slavery, Caste System, Labor Allocation, ...)	Yields added to classes of improvements	The labour regime (section 8): which occupations exist and their output
Taxation (Tribute, Census, National Debt, ...)	Capital and yield modifiers	The share of commerce taxed, tariffs on foreign goods, debt
Trade policy (inside Economy today)	National Socialism forbids foreign trade	Free trade, mercantilism (tariffs, colonies), autarky (embargo)

The starting civic Oral Law shows why civics must become levers on the ledger rather than percentages: its –25 % food, applied to the harvest, costs a quarter of the population by turn 150 (section 6.3).

13. Trade

Inside a civilization goods move freely within a network (roads, rivers, harbours connect it); between a civilization's separate networks they move by convoy. **Between civilizations** they move by convoys along trade routes, under contracts or as spot sales. Every quantity offered, asked or delivered is a whole number of tons.

Instrument	What it is	Engine
Spot sale	Tons of a good sold now at the partner's market price	One deal item: good and tons
Contract	Tons per turn, for a term, at a price; penalties for default; cancelled by war	One deal item per good, repeated each turn of the term

Instrument	What it is	Engine
Seeds, breeding stock, techniques	The means to produce a good: the buyer can plant, breed or build it	A deal item like a technology
Package	Any combination with gold, technologies, treaties	The existing deal structure
Tariffs, embargo, blockade	A share of value taken at the border; no trade; ships cut a sea route	Civics; diplomacy; naval units on route plots

A deal item is a type and one 32-bit integer (`TradeData`, CvStructs.h, exported to the executable). A goods item packs the good and the tons into that integer, d = tons × N + good for N goods, and refuses more tons than fit. With N = 64, one item carries up to 33,554,431 t a turn; more takes a second item. Whether the executable's own trade screen reads the integer of a resource item directly is not known from the DLL source (*open*; the first test of milestone M4).

A trade route becomes a convoy's capacity: it carries goods worth its capacity per turn and earns the price difference. In the 30 games, all trade that could happen between players needed 0.42 units per route per turn with one route per city, so routes become a real constraint only as trade grows, which is the intent: harbours, roads and ships matter because they carry goods. Food travels well only once it keeps: before Fermentation a convoy of fish loses a fifth of its load each turn on the road.

The AI trades by value. A deal is worth the sum of the marginal values (section 16) of what is received minus what is given, including the strategic value of giving a rival the means to produce. This replaces the random offers and fixed resource values that the diplomacy work of merge request !63 began to remove.

14. War

Units are built from production and goods (a legion needs iron arms; a ship of the line timber, cannon and canvas; a tank steel, fuel and ammunition). Armies in the field eat 300 t per unit and turn from the nearest connected city, and their food spoils on the road like any other; a unit cut off loses strength. Battles consume ammunition from the industrial era on. Blockades and raids on provinces become ways to win: a country without steel cannot replace its tanks, and a besieged city lives on its box. An arsenal stores arms in peace; the model showed a 90-turn war emptying a 60-unit arsenal by its eleventh turn (Land, not tokens, section 9).

15. The eras

Era	The economy's question	What changes
Ancient	Can we keep what we harvest, and clothe ourselves?	Pottery and Fermentation end most spoilage; everyday goods; bronze needs copper and tin, rarely found together
Classical	Who controls iron and the roads?	Iron tools raise all work; coinage creates markets; roads join networks
Medieval	Can the towns grow beyond the land?	Guild workshops; mills (power from water and wind); estates and serfdom
Renaissance	Who controls the sea and the colonial goods?	Comfort goods from the tropics; banks; mercantilist trade
Industrial	Coal, iron and the factory	Workers run several recipes; power; railways move goods; Refrigeration ends spoilage; cities outgrow their farms
Modern	Oil and the consumer	Oil and electricity; consumer goods raise living standards; global supply chains
Transhuman	Automation	Androids (RoM's specialist) and automated factories: work stops being people
Future	The end of scarcity	Replicators make goods from energy; the ledger's constraints relax by design

A Normal game lasts 700 turns (Marathon 1,470); the 30 measured games cover its first 150, from 6000 BC to 1300 BC. Each era's goods should become important within its first third and remain useful after: the design target is that no good introduced in an era stops mattering before the era after next.

16. The AI

The AI uses one quantity for every economic decision: the marginal value of one more ton per turn of each good to each of its networks, the change in its weighted living standards and output when that supply rises:

$$v_g = V(\mathbf{q} + \mathbf{e}_g) - V(\mathbf{q}), \qquad V = \sum_{cities} \left(\sum_{tier} \omega_{tier} k_{tier} + \omega_B B + \omega_F F + \omega_P P + \omega_C C \right) \tag{10}$$

k is the citizens supplied in each tier (section 10) and *B* the food box, so food saved from spoilage has a value like food grown, and Pottery, Fermentation and Refrigeration are researched for the tons they save. Computed by recomputing the network once per good, it costs 98 ms per turn at 300 cities and 100 goods on the measured host, against a late turn of about 20.8 s today (section 19.5). It drives: occupations and plots (the governor); what to build (recipes whose outputs are worth more than their inputs and labour); what to research, plant and prospect; what to offer, ask and accept in trade; when stores are worth filling; and which provinces and routes are worth a war. Because every decision uses the same values, the AI cannot be inconsistent across them, which is the failure mode of today's separate fixed weights.

17. The interface

- **City ledger screen** (Python): every good made, used, stored, received and spoiled, in tons; each tier as "8 of 10 citizens clothed", with what is missing and where it could come from.
- **Food box**: the tons held, the threshold, the tons lost to spoilage this turn and the technology that would stop them.
- **Land overlays**: crop suitability, ore grade and depth, fishing grounds, climate; on the map through coloured plots, the API the strategy overlay already uses.
- **Market screen**: the network's prices and their history; what is short; what sells.
- **Contracts and convoys**: tons in and out, with whom, until when.
- **Plot and city help**: every number names its cause (this plot, this worker, this building, this civic).
- **Advisors**: shortages and opportunities in one line each: "Iron town: 600 t of grain short for 3 turns; the road to the farms is cut".

18. Numbers and where they come from

18.1 Anchors from the engine

Quantity	Value	Source
Food per citizen	3 a turn, so 300 t	FOOD_CONSUMPTION_PER_POPULATION (GlobalDefines.xml)
Growth threshold	(30 + 10 × size) × 100 t	BASE_CITY_GROWTH_THRESHOLD, CITY_GROWTH_MULTIPLIER; CvPlayer::getGrowthThreshold
Hunger	box below zero: emptied, one citizen lost	CvCity::doGrowth
Box kept after growth	Settlement 20 %, Town and City 30 %, Metropolitan 40 % Essentials	iFoodKept, CIV4BuildingInfos.xml
Pottery, Fermentation, Refrigeration	cost 60, 95, 5,200; 2, 5, 88 technologies at least	CIV4TechInfos.xml
Plots per city	21	NUM_CITY_PLOTS
Game length	700 turns Normal, 1,470 Marathon	CIV4GameSpeedInfo.xml
Yields and commerce outputs shown	3 and 4	YieldTypes and CommerceTypes, compiled into the executable
Deal item	a type and one 32-bit integer	TradeData (CvStructs.h), exported to the executable
Specialist types today	19	CIV4SpecialistInfos.xml

18.2 Measured in the 30 games

Quantity	Value	Source
Rebuilt cities fed by their plots	93.1 % of city-turns	Settlement Economy, section 4
Production over plots, turn 150	4.08 (median player)	Settlement Economy, section 5.3
Trade routes at the 1.00 floor, turn 150	2708 of 3025	Settlement Economy, section 5.4
Hunger under the food rule	0.9 to 3.3 per 10,000 city-turns	Section 6.3
Population lost without Pottery, turn 150	16 %	Section 6.3
Food held at the threshold	11.2 % of the harvest	Section 6.3
Allocations changed by floating-point order	1803 of 120323	Settlement Economy, section 7.5
Resource deals between AIs that paid gold	1 of 580	Settlement Economy, section 5.1
Cost per turn, 300 cities and 100 goods	0.95 ms; 98 ms with marginal values	Settlement Economy, section 7.8

18.3 Need levels solved from targets

The targets are design intents: everyday goods for 8 of 10 citizens, comforts for 5 of 10. Under the recommended rules (goods shared in the network, stores, spoilage, a governor that knows needs; levels by development, a town 1.4 and a city 2.0 times a village), a bisection over one level per need, run on all 30 games each step, gives the levels of section 10. With them, 81.1 % of citizens have everyday food variety, 80.0 % clothing, 80.3 % fuel and 49.8 % comforts. These runs took each city's size from the games (the observed population split among a player's cities), so they measure supply, not growth; growth is measured by the food rule of section 6.

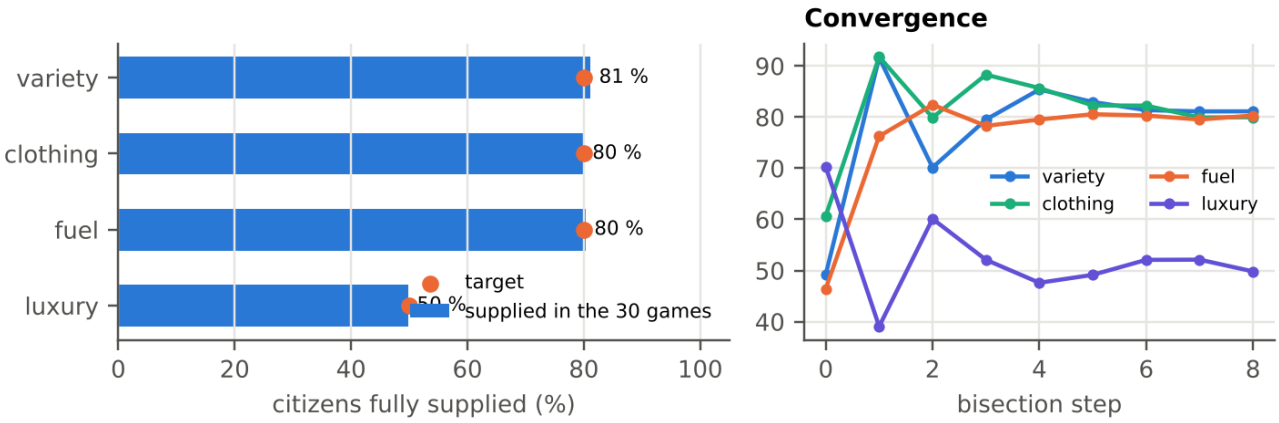


Figure 4. Left: the share of citizens fully supplied with each need at the solved levels, against its target. Right: how the bisection converged.

18.4 Design values and their levers

The spoilage rates of section 6.2 are design values: their order follows how fast each food spoils (fish, then meat and dairy, then fruit, then grain), and their size was set so that missing Pottery costs 16 % of turn-150 population and knowing all three costs nothing. Figure 3 (right) is the lever: the measured cost at every scale from none to three times these rates. These levels hold for the land alone in the first 150 turns of one scenario. Each later era adds workshops and imports to the supply and must be solved the same way, with the scenario set widened to the map types RoM ships.

19. Architecture

Executable (unchanged)	City screen: 3 yields, 4 commerce sliders	Diplomacy screen: deal items	Map: improvements, units, overlays
Python (new screens)	City ledger screen	Market and contracts screen	Land overlays, plot help
DLL (new and changed)	CvLedger, CvNetworkEconomy	CvMarket, CvContract	Yields as views, AI valuation
XML (data)	CIV4GoodInfos, needs by tier	Recipes on buildings, specialists	Goods costs on units, civic levers

Figure 5. Where each part lives. The executable is not changed; everything new is DLL, XML and Python.

19.1 New classes

Class	Holds	Saved
CvGoodInfo (XML: CIV4GoodInfos.xml)	Tier, spoilage per technology, storage, base price, elasticity, the resource type it reuses for art and text	no
CvLedger (in CvCity)	Per good: stock, made, used, received, sent, spoiled (int tons)	yes, behind CvCity's uiFlag
CvNetworkEconomy (per CvPlotGroup)	Recipes in dependency order, rationing, transport capacity, the market	derived each turn
CvMarket	Prices and their history per good	yes
CvContract (per player pair, in CvDeal)	Good, tons per turn, term, price, penalty	yes

19.2 Changed functions

- `CvPlayer::doTurn`: the economy pass (figure 6) before the loop over cities.
- `CvCity::getBaseYieldRate`: food, production and commerce as views of the ledger (equations 2 to 4).
- `CvCity::doGrowth`: the box in tons, eaten perishable first, spoiling by the owner's technologies; growth, kept share and hunger unchanged.
- `CvPlot::calculateYield` and improvement yields: from land qualities, not resources.
- `CvCity::changeFreeBonus`: the bridge that lets resource-type requirements read the ledger during the transition.
- `CvDeal`: goods items in tons and contracts; `CvPlayerAI`, `CvCityAI`: the valuation engine.
- Map scripts and WorldBuilder loading: provinces from the old resource plots, then the plots cleared.

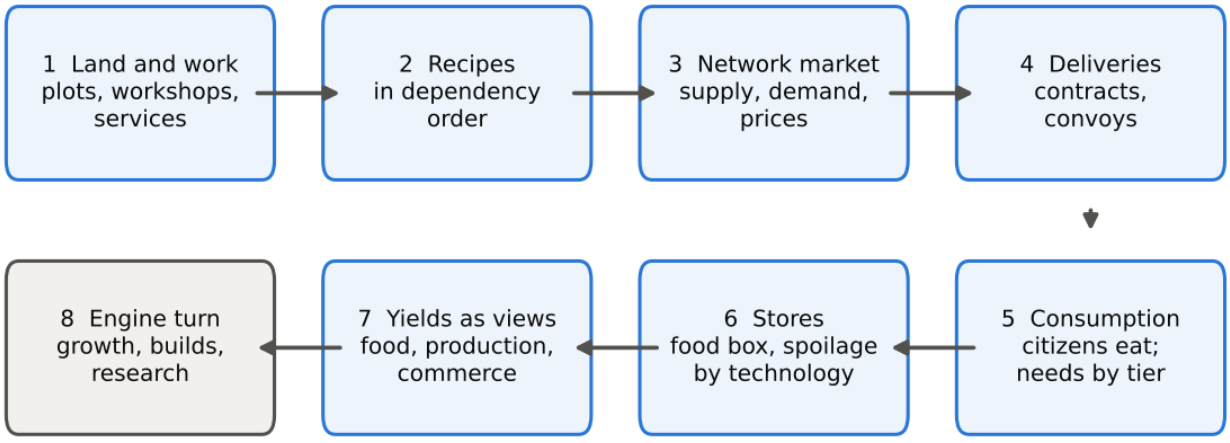


Figure 6. One economy turn, per trade network, before the engine's own turn.

19.3 Integers that cannot overflow

Every quantity is an integer number of tons, and its width is chosen from a bound taken from the records, not from what the games happened to reach:

- **One city, one good: 32-bit.** The most food any city could harvest in a turn, with the best plot of every kind on all 21 plots and every food building and civic at once, is 2,141 food, 214,100 t (the bound is far above any real city). Size is limited by food to 713, whose threshold is 716,000 t, so a box never exceeds 930,100 t: 2,308 times below the 32-bit limit of 2,147,483,647. In the 30 games the largest harvest was 9,590 t and the largest box 27,040 t.
- **Stores are capped.** Every stock is held to its storage C_g (equation 1); a delivery beyond it is refused at the sender, so nothing overflows and nothing vanishes.

- **Sums and products: 64-bit.** Totals over cities, players and the world, and every product (price × tons, tons × rate) are computed in 64-bit integers (`__int64`, which the MSVC 2003 compiler of the DLL supports), dividing last.
- **Deal items** carry at most 33,554,431 t each (section 13); a larger deal takes more items.

19.4 Determinism and saves

All rules are integer arithmetic in tons, rationing is exact (largest remainders, ties by index), and every random draw (harvests) uses the synchronised generator; the growth runs of section 6 gave the same result twice. New data sits behind the version flags the classes already read first. Old saves load with empty ledgers and their food boxes converted at 100 t per unit.

19.5 Performance budget

Measured in C on the study host: 0.95 ms per turn for the goods pass at 300 cities and 100 goods, 98 ms with the AI's marginal values; at 600 cities and 40 goods, 0.91 ms and 37 ms. The budget for the whole economy is 1 % of a turn; at today's late-turn time of about 20.8 s that is 200 ms.

20. Building it

The Living Economy is built on one branch, as a whole, not merged piece by piece into today's economy. The branch is playable at every milestone because the bridge (CvCity::changeFreeBonus) keeps every unconverted rule reading goods as resources.

Milestone	Content	Done when
M1 Ledger, land and food	Goods XML; CvLedger in tons; land qualities; plots make goods; the food box in tons with spoilage by technology; the three yields as views; the bridge	Harness: no crash or slower turns (5 seeds); hunger no more frequent than today's rule; population at turn 150 within the bracket of section 6.3
M2 People and living standards	Occupations; workshops; tiers counted in whole citizens; the needs-aware governor; the ledger screen	Harness: 8 of 10 citizens with everyday goods and 5 of 10 with comforts, within 3 points (30 seeds)
M3 Markets and money	Network markets, prices, commerce from sales; civics as levers	Harness: commerce per citizen within 15 % of today's by era
M4 Trade	Goods deal items in tons, convoys, contracts, seeds, the AI's deal valuation; the Python trade screen	Harness: AI trade volume and its balance; the executable's trade screen shows goods items; the user's games
M5 Industry and war	Recipes with tools and power; units costing goods; supply; ammunition	Harness: war outcomes against fixed opponents (30 seeds)
M6 Later eras and balance	Industrial, modern and future goods; need levels per era solved again	The user's games through the late game

Each milestone is a registered prediction judged under the repository's rules. The deterministic models are extended with each milestone first, so every rule is measured on real cities before it is written in the DLL.

21. Risks

Risk	How the design answers it
The AI cannot play it	One valuation engine for every decision; measured affordable; each milestone judged on AI outcomes first
Too much to manage	Distribution is automatic; governors assign work; contracts run themselves; the player sets direction
Spoilage stunts the early game	Pottery is one technology from every start; its absence costs 16 % and its earliest arrival 2.0 % of turn-150 population; the rates are levers with a measured curve
Balance collapses when one part changes	Need levels and prices are solved from targets, not tuned by hand; the model re-solves them for every change
Numbers overflow or drift	Whole tons in integers sized from bounds in the records; 64-bit sums; exact rationing; floating point was shown to change allocations
The executable's screens cannot show it	Three yields and four commerce outputs kept as views; new information in Python screens; goods deal items tested first in M4
Old saves and scenarios	Provinces from old resource plots; version flags; empty ledgers on load
Later eras untested	M6 re-solves them; the 30 games cover only the first 150 of 700 turns, and no player reached Refrigeration